Code No. 3525

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (CSE, AI&ML) V-Semester (MAIN) Examinations Dec 2023 /Jan 2024
DATABASE MANAGEMENT SYSTEMS

Time : 3 hours **Max. Marks : 70**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7).*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

		Marks	BT	CO
1. a)	Define Database. Discuss about applications of Database Systems.	2	2	1
b)	What is a weak entity. Explain with example.	2	2	1
c)	When can we use group by clause. Explain.	2	3	2
d)	Discuss about super key and candidate key in functional dependency with example.	2	2	3
e)	Define transaction management.	2	1	4
f)	Define validation-based protocols.	2	2	5
g)	Give an example for Extendible hashing.	2	2	4
2. a)	Draw the Architecture of Database. Explain about different views of data.	7	4	1
b)	Write about logical database design (ER to Relational) with suitable examples.	7	4	1
3. a)	Explain in detail about nested queries.	7	3	2
b)	Discuss about outer joins with examples.	7	4	2
4. a)	Define normalization. List and explain different normal forms with examples.	10	4	3
b)	What is meant by integrity constraint. Write about complex integrity constraints in sql.	4	3	3

(P.T.O.)

5. a)	Explain about B+ Trees Dynamic Indexing.	6	3	4
b)	Explain serializability in transaction management.	8	4	4
6. a)	Explain concurrency control with lock-based protocols.	7	3	5
b)	Explain buffer management in concurrency control system.	7	4	5
7. a)	Write short notes on Variations in relational calculus	7	4	2
b)	Differentiate Static hashing and Dynamic hashing.	7	3	4

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